

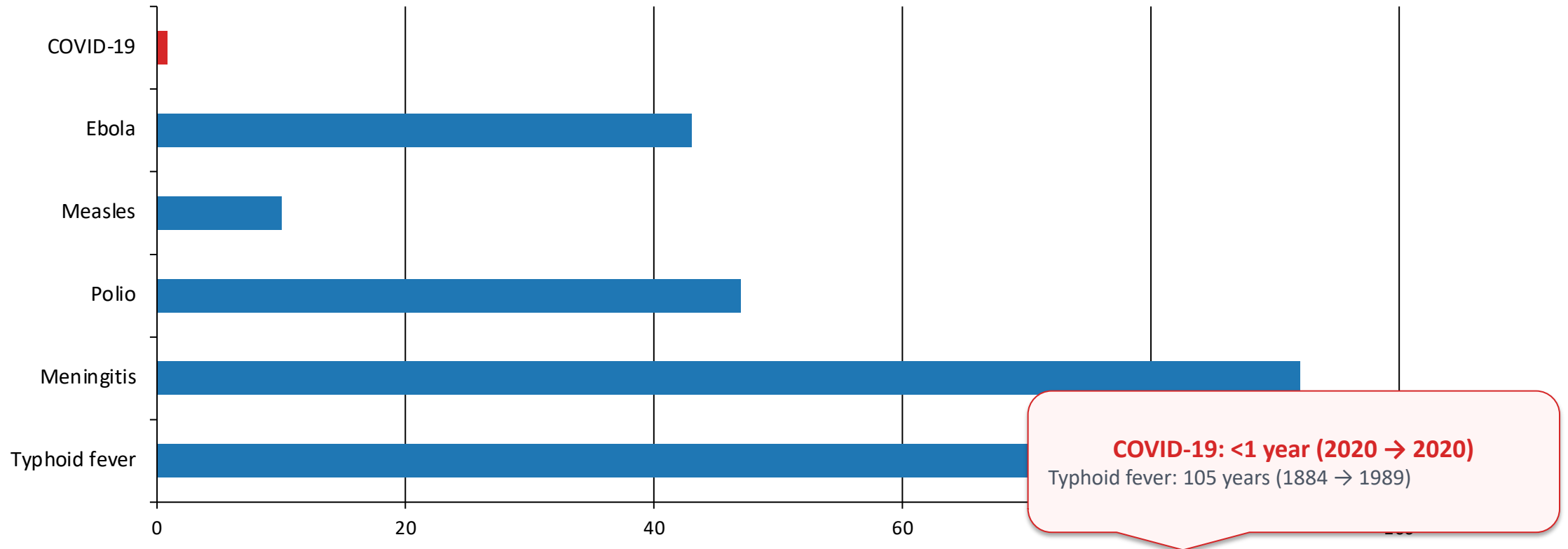
The Power of Vaccination: From Smallpox Eradication

Historical trajectory, quantified impact, and remaining coverage gaps (1796–2020)



From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Years from pathogen identified to vaccine licensed



US IMPACT

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox

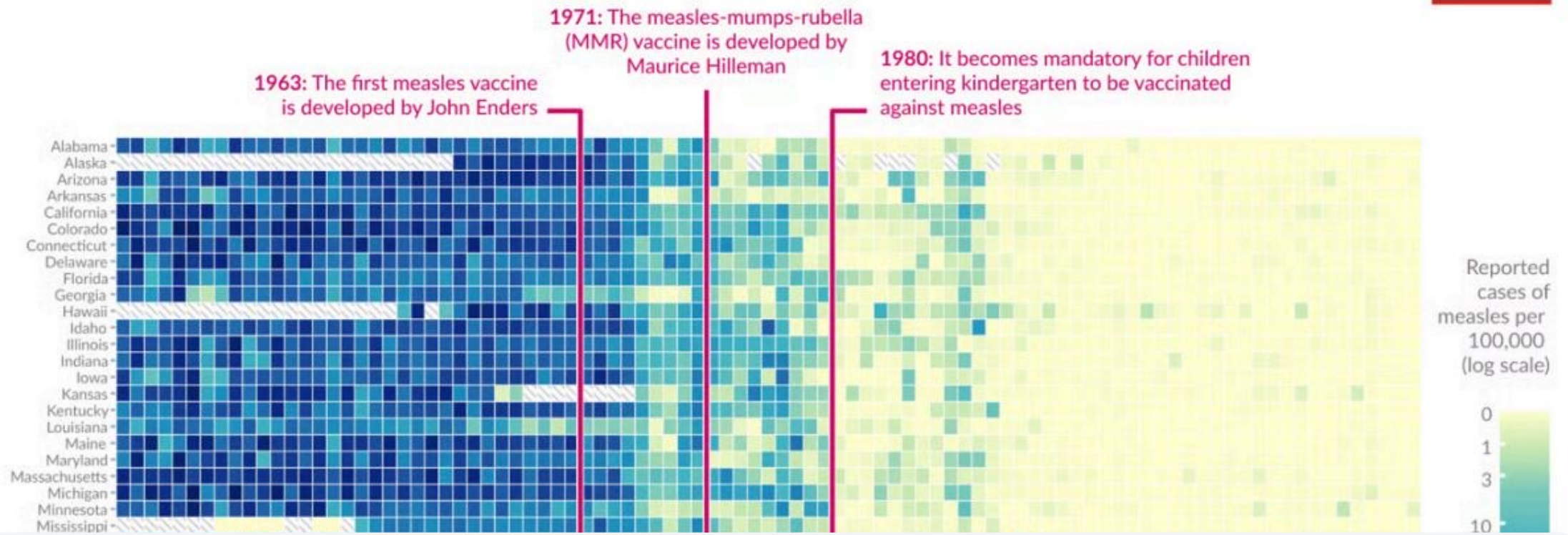
Disease	Pre-vaccine cases (per million/yr)	Post-vaccine cases (per million/yr)	Case reduction	Pre-vaccine deaths (per million/yr)	Death reduction
Diphtheria	158	0	100% ✓	13.7	100%
Smallpox	250	0	100% ✓	2.9	100%
Acute polio	141	0	100% ✓	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Pneumococcal	233	139	40.5%	24	31.3%
Acute hepatitis B	273	44	83.9%	1	83.6%

Table values are per million population per year (Roush & Murphy, 2007; via Our World in Data).

Measles Cases Collapsed Across All 50 US States After Vaccine M

Vaccines reduced measles cases across US states

Our World
in Data



Milestones: 1963 first measles vaccine • 1971 MMR vaccine • 1980 mandatory for kindergarten entry

ERADICATION

Smallpox: The Only Human Disease Eradicated Through Vaccina

Timeline

1959: WHO resolution to eradicate smallpox
1967: Intensified global campaign launched
1977: Last natural case (Somalia)
1980: Eradication declared worldwide

Why eradication was possible

Infected only humans
Distinct, recognizable symptoms
Vaccine was cheap and effective
Endemic countries were tracked and targeted

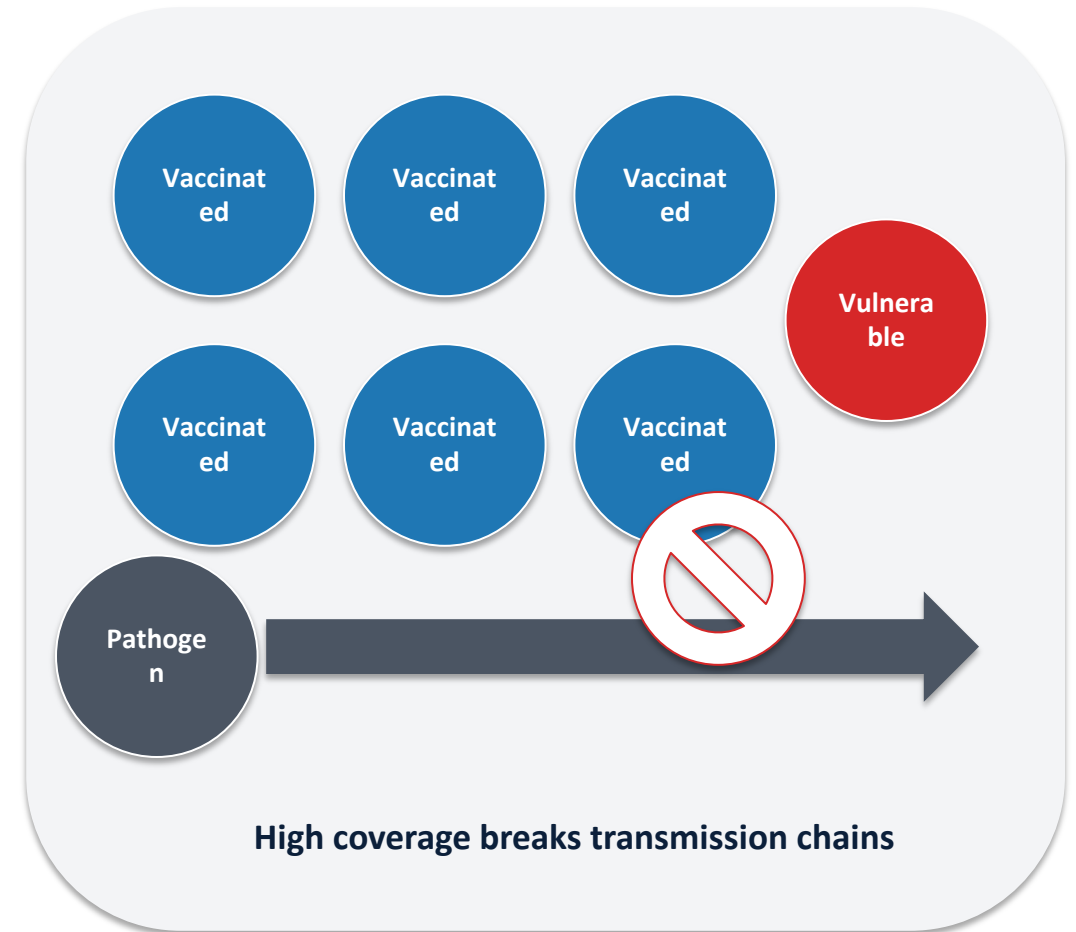
MECHANISM

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Vaccines protect individuals and reduce spread at the population level.

When enough people are vaccinated, pathogens cannot spread effectively.

This helps shield people who cannot be vaccinated: newborns, immunocompromised individuals, and cancer patients on chemotherapy.



REMAINING CHALLENGES

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Redu

Incomplete impact (US reductions)

Pneumococcal: 40.5% case reduction; 31.3% death reduction

Hepatitis A: 89% case reduction

Acute hepatitis B: 83.9% case reduction

Global access remains unequal.

Supply shortages can put children at risk.

Poor scientific understanding can reduce trust and uptake.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — B

Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and acute polio in the US.

Development timelines collapsed from 105 years (typhoid) to <1 year (COVID-19).

Measles dropped to near zero across all 50 states after the 1980 kindergarten mandate (with sporadic outbreaks).

Herd immunity protects whole communities, including people who cannot be vaccinated.

Gaps persist for pneumococcal disease (40.5% case reduction) and hepatitis A/B—requiring improved access, coverage, and trust.

Call to action: Strengthen vaccination systems to expand access, maintain high coverage, and build trust.